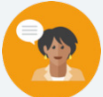
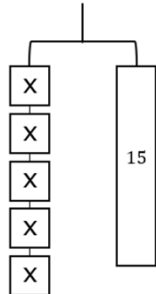


14.6 Solving One-Step Multiplication and Division Equations in One Variable Using Models

Lesson Introduction

 Benchmark	MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.		 Learning Targets	- I can solve one-step multiplication and division equations using manipulatives. - I can solve one-step multiplication and division equations using diagrams.
 Benchmark Coherence	Building On MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.		Working Toward MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- How would you use algebra tiles to model and solve the multiplication and division equations? - How do you use tape diagrams to model and solve one-step multiplication and division equations? - How do you use hanger diagrams to model and solve one-step multiplication and division equations? - How do you solve multiplication and division one-step equations?			
 Lesson Narrative	In this lesson, students use manipulatives, tape diagrams, and algebra tiles to solve multiplication and division one-step equations using inverse operations. Students use inverse operations to isolate the variable while making sense of the meaning of the variable and the operation using models.			
 Component Summary	14.6.1 Warm-Up (~5 min.)	Error analysis of image (<i>SE p. 367</i>)	14.6.3 Guided Instruction (10-15 min.)	Learn how to model and solve division one-step equations (<i>SE p. 370</i>)
	14.6.2 Exploration (10-15 min.)	Explore how to model and solve multiplication one-step equations (<i>SE p. 368</i>)	14.6.4 Collaboration (10-15 min.)	Match one-step multiplication and division equations and solutions (<i>SE p. 372</i>)
	14.6.6 Wrap-Up (~5 min.)	Write a sales pitch for inverse operations (<i>SE p. 373</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Inverse operations		- Algebra tiles - Colored pencils (Optional) Chart paper (Optional) Markers	- Consider using virtual manipulatives. - Consider putting algebra tiles in zip top bags with the contents labeled for students to use or share. - If you do not have algebra tiles, you can create them using a template or use virtual algebra tiles.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may think that the coefficient represents the tens place and the variable represents the ones place (of a two-digit number). - Students may think the inverse operation of a negative coefficient is to add the coefficient to both sides (instead of dividing by it). - Students may not perform the inverse operation to both sides of the equation. - Students may struggle with the models for division equations.	- Saying “2 multiplied by x” instead of “$2x$” will help students remember that the coefficient is multiplied by the variable. - If time is an issue consider assigning Your Turn! as homework. - The models for division equations may be difficult for students. Consider have strips of paper that students can divide up into parts. For example, for 14.6.3 Guided Instruction question 1b provide the students with a strip of paper that is partitioned into 3 parts. Students can write $\frac{x}{3}$ on each part and cut the strip. Then the student can use the paper with $\frac{x}{3}$ to set equal to 4 and repeat that to demonstrate the model in question 1b.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 14.6.4 Collaboration, students explain the rationale for their matches and listen to their partner's rationales, taking turns deciding, explaining, and listening. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.	MA.K12.MTR.2.1: Multiple Representations In 14.6.2 Exploration, students learn how to model and solve multiplication one-step equations using algebra tiles. In 14.6.3 Guided Instruction, students learn how to model and solve division one-step equations using algebra tiles. After learning how to represent and solve one-step equations using algebra tiles, students practice in the subsequent components.
 Differentiate Instruction	Visual, auditory, and kinesthetic entry points are built in throughout this lesson. Hanger diagrams can be physical or digital manipulatives that benefit students who learn best with hands-on resources.	
Provide English Language Learners with anchors of support, such as vocabulary cards of the words used in this lesson: equation, positive, negative, inverse operations, integers, solutions.		
Lesson Notes:		

14.6.1 Warm-Up: Math Fail

Component Overview: Students analyze an image to determine the “math fail.”



14.6.1 Warm-Up: Math Fail

This children’s book has published a “math fail.”

1. Explain the error shown in this image.



There are 6 bananas in the image, but the writing shows there should be five.



Consider an extension for homework in which students have to find “math fails” in their world—at the grocery store, on a sale sign, or in a book.

14.6.2 Exploration: Exploring Relationships

Component Overview: Students explore how to model and solve multiplication one-step equations using algebra tiles. Students develop a conceptual understanding of equations as balanced scales. This helps students understand why they need to perform inverse operations to both sides of the equation. Finally, students review the inverse operations for addition, subtraction, multiplication, and division.

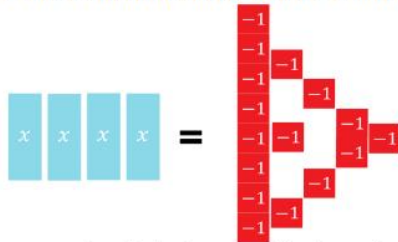


14.6.2 Exploration: Exploring Relationships

Recall that when solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

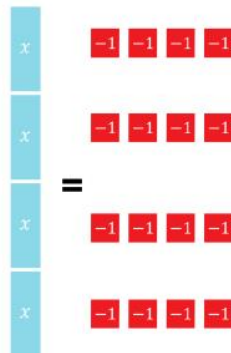
1. A model of the equation $4x = -16$ is shown using algebra tiles.

The  tile represents the variable, x , and the  represents the constant term where each tile is 1 unit.



- a. What operation is being applied to the 4 and the x in the equation $4x = -16$?
Multiplication

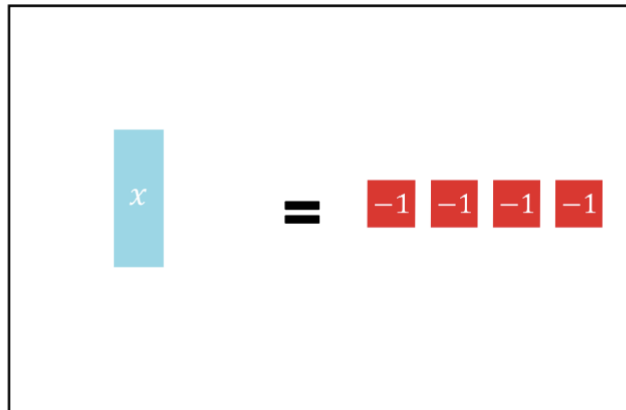
- b. The algebra tiles are arranged on the equation mat into equal-sized groups on both sides of the equation. How many groups were created?
4 groups



Ask students how they think the variable can be isolated. This could be rephrased as grouping each side to show how many each x represents. Have students share their ideas and then test them using the algebra tiles. Students may be able to figure out how to solve the equation independently if given time to experiment with the algebra tiles.

14.6.2 Exploration: Exploring Relationships (continued)

- c. Use algebra tiles to sketch the solution of the equation.



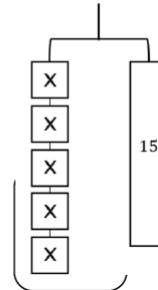
This section provides a variety of entry points through the visual representations of the hanger diagrams and partner discussions. Consider providing algebra tiles for a kinesthetic entry point and encouraging students to model the equation first using tiles if they are struggling with the hanger diagram representation.

- d. What is the value of x ?

$$x = -4$$

2. The hanger diagram represents the equation $5x = 15$.

- a. Complete the statements.



The left side of the hanger is broken into equal parts. If 15 is broken into the same number of equal parts, each part will have the same weight as one x block. Splitting each side of the hanger into

- ☐ 3
- ☒ 5
- ☐ 15

the same as

- ☒ dividing
- ☐ multiplying

each

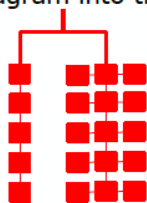
side of the equation by

- ☐ 3.
- ☒ 5.
- ☐ 15.

14.6.2 Exploration: Exploring Relationships (continued)

- b. Sketch splitting the hanger diagram into the equal parts as described in Part A.

Answers will vary.



- c. What is the value of x ?

$$x = 3$$

- d. Compare your diagram with your partner. Discuss the similarities and differences.

Answers will vary.

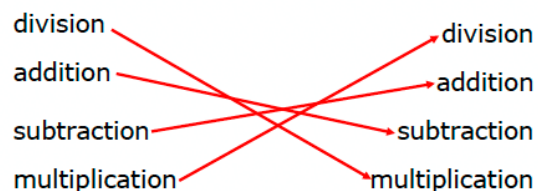
- e. Provide positive feedback to your partner in their book on their sketch and solution.

Answers will vary.

**Partner
Feedback**

3. Inverse operations can be helpful in solving equations.

- a. Draw an arrow to match each operation on the left with its inverse operation on the right.



- b. Explain what it means for operations to be inverses.

Answers will vary. Inverse operations "undo" each other.

- c. Which inverse operations were used to solve the equations in this exploration?

Multiplication and division



Model what positive feedback looks and sounds like prior to this activity with examples. Consider providing a posted anchor chart that students could reference.



If students are struggling to match, explain inverse operations by further scaffolding question 3 through questioning:

- *What does "inverse" mean?*
- *How can you determine the inverse of an operation?*

14.6.3 Guided Instruction: Using Models to Solve Equations

Component Overview: Students model and solve division one-step equations using algebra tiles. Students solve the equation by multiplying both sides of the equation by the number the variable was divided by. Remind students that they can check their solution by substituting it in for the variable and seeing if the left side equals the right side.



14.6.3 Guided Instruction: Using Models to Solve Equations

Algebra tiles can also be used to model division equations.

1. The equation $\frac{x}{3} = 4$ can be rewritten as $x \div 3 = 4$.

To represent division, show that x tile is being divided into equal parts. Sketch lines on the x tile to represent the fraction $x \div 3$.

This means when x is divided into 3 parts, each part is equal to 4. The model shown illustrates that $\frac{x}{3} = 4$.

$$\frac{x}{3} = 4$$

- a. If each $\frac{1}{3}$ of x is equal to 4, discuss with your partner what the value of x is. As part of your discussion, include which operation can be used to determine the value of x . Summarize your discussion.

Answers will vary. x is equal to 12. The problem is using division so the inverse that would need to be used is multiplication.



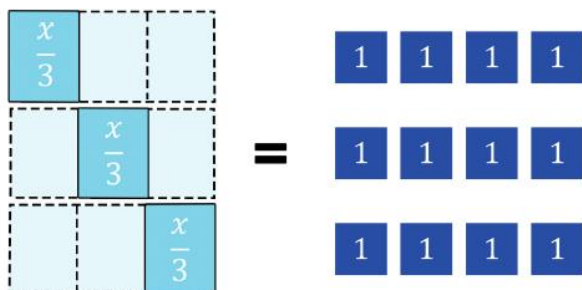
Consider creating an anchor chart with instructions, examples, and visual models for students to reference in subsequent components.



Students could say addition or multiplication. If students say addition, it would be repeated addition, which can be related to multiplication.

14.6.3 Guided Instruction: Using Models to Solve Equations (continued)

- b. The image models the relationship discussed that can be used to determine the value of x .



Explain how this representation can be used to determine the value of x .

Answers will vary.

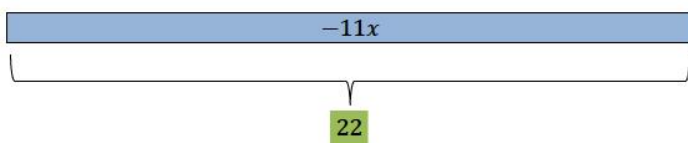
One-third of the boxes is four.

- c. What is the value of x ?
 $x = 12$
- d. Complete the statements by writing operations on each blank.

The original equation involved the operation Division. Its inverse operation, Multiplication, was used to solve the equation.

Tape diagrams, also known as bar models, can be used to model and solve multiplication and division equations.

2. The tape diagram models the equation $22 = -11x$.



Take the time to explain the model in question 1b. Point out how each $\frac{x}{3} = 4$. Then, help students see that the left side is three $\frac{x}{3}$'s which is equivalent to x , and since the left side has three $\frac{x}{3}$'s, the right side needs three sets of the four one-unit tiles. Lead students so that they relate the model to multiplication.



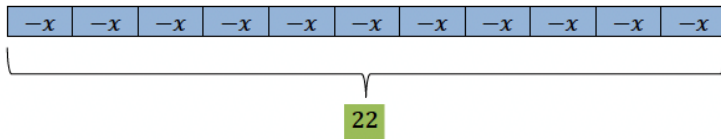
To provide additional scaffolding, ask guiding questions like:

- What is the inverse operation of division?
- What is the inverse operation of multiplication?
- Why do you perform the inverse operations on both sides of the equation?
- How can you check your equation to make sure the solution is correct?

14.6.3 Guided Instruction: Using Models to Solve Equations (continued)

The solution to the equation $22 = -11x$ can be modeled in the following steps. Work with a different partner to explain what is occurring in each step.

Step 1



- a. How were the integers in the equation used to determine the negative value of x ?

Explanation of Step 1

Answers will vary. $-11x$ has been broken up into 11 individual pieces with each piece equivalent to $-x$.

Step 2



Explanation of Step 2

Answers will vary. The value of $-x$ is 2.

Step 3



Explanation of Step 3

Answers will vary. We want to find the value of x , so we take the opposite $-x$ and the opposite of 2, which gives us -2 .

- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.
Answers will vary.



Consider providing sentence stems for students to discuss their responses and explanations to provide additional entry points for learners.

"The operation used in the equation was _____. I solved the equation by using _____. This worked because..."



If students are able to solve the equations without visual models, allow them to do so. Ask them to check their solutions. If they finish early, consider having them write real-world word problems that could be modeled by the equations in the lesson.



To encourage mathematical reasoning, ask questions like:

- Does this answer make sense?
- Did you check your solution by substituting it back into the original equation?
- How can you use mental math to help you find the correct diagram that models the equation?

14.6.4 Collaboration: Matching Equations and Solutions

Component Overview: Students match one-step multiplication and division equations and solutions. Then students explain how they determined which equations to match with which solutions to their partners.



14.6.4 Collaboration: Matching Equations and Solutions

- Match each equation to the correct solution. Record the equation with its solution in the table. Some solutions may be used more than once or not at all. Include an explanation for the match.

A. $x = 2$	E. $x = 3$
B. $x = -20$	F. $x = 10$
C. $x = -10$	G. $x = -8$
D. $x = 9$	H. $x = -3$

- $-3x = -6$
- $-\frac{x}{2} = 4$
- $\frac{x}{3} = 3$
- $-\frac{x}{2} = 5$
- $3x = 9$
- $\frac{x}{10} = -2$
- $4x = 8$

Equation	Solution	Explanation for Match
1	A	Answers will vary. When substituting 2 in for x in $-3x = -6$ the result is -6 .
2	G	Answers will vary. When substituting -8 in for x in $-\frac{x}{2} = 4$ the result is 4.
3	D	Answers will vary. When substituting 9 in for x in $\frac{x}{3} = 3$ the result is 3.
4	C	Answers will vary. When substituting -10 in for x in $-\frac{x}{2} = 5$ the result is 5.
5	E	Answers will vary. When substituting 3 in for x in $3x = 9$ the result is 9.
6	B	Answers will vary. When substituting -20 in for x in $\frac{x}{10} = -2$ the result is -2 .
7	A	Answers will vary. When substituting 2 in for x in $4x = 8$ the result is 8.



Take Turns

Students explain the rationale for their matches and listen to their partner's rationales, taking turns deciding, explaining, and listening. Taking turns can also give students more opportunities to construct logical arguments and critique others' reasonings.

14.6.5 Wrap-Up: Ticket Out the Door

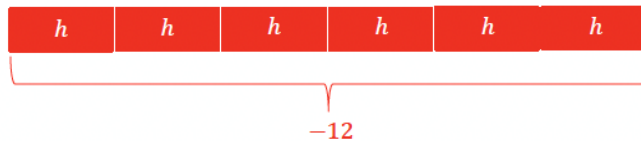
Component Overview: Students use a visual model to solve a one-step multiplication equation. Use data from this formative assessment to determine which students need remediation before the next lesson.



14.6.5 Wrap-Up: Ticket Out the Door

- Throughout this unit, you have been shown how to solve equations using tape diagrams, hanger diagrams, number lines, manipulatives, and inverse operations. Use one of these methods to model and solve the equation $6h = -12$.

Diagram answers will vary. $h = -2$



This section allows for multiple entry points, as students have a choice for how they want to represent and solve the equation. Encourage students to choose the method that makes the most sense to them.